Question Paper Code 25901291

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION – March 2026
Fourth SEMESTER
Discipline Specific Elective (DSE) - MG4DSEGEO200 - ENVIRONMENTAL GEOLOGY
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

ANSWER KEY

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Short Answer in 60 words
Answer any 5 questions
Each question carries 2 marks

1. Definition (2 marks). [U]/[CO1]
2. Protection of resources from human exploitation to maintain them in their original state (1 mark); National parks, wildlife sanctuaries, or biosphere reserves (Any example-1 mark). [U]/[CO2]
3. Desertification, loss of soil fertility, permanent depletion of groundwater aquifers, land subsidence. (Any 2 impacts-2 marks). [K]/[CO3]
4. Established that all men are created equal with unalienable rights, shifted the source of rights to the individual (Any other details-2 marks). [K]/[CO5]
5. Non-dominant groups possessing stable ethnic, religious, or linguistic characteristics different from the majority (1 mark); Examples in India: Muslims, Christians, Sikhs, Buddhists, Jains. (Any 1 example -1 mark). [K]/[CO5]
6. The Water (Prevention and Control) Act, 1974, The Environment (Protection) Act, 1986. (Any 2 laws-2 mark). [U]/[CO1]

[2×5=10]

Part B
Short Notes in 250 words
Answer any 6 questions
Each question carries 8 marks

7. Definition and mechanism (5 marks); Ozone depleting substances explanation and examples (3 marks). [U]/[CO2]
8. Environmental protection (4 marks); Social equity (2 marks); Economic viability (2 marks). [U]/[CO3]
9. Early concepts (Magna Carta, Bill of Rights) (2 marks); 8th-19th Century (American/French Declarations) (2 marks); 20th century milestones (League of Nations, UDHR 1948). (4 marks). [U]/[CO5]

10. Art 14: Equality before law. (3 marks); Art 15-16: Prohibition of discrimination and equality of opportunity in employment. (3 marks); Art 17-18: Abolition of Untouchability and Titles. (2 marks). [A]/[CO5]
11. Hydrosphere- explanation (4 marks); Hydrological cycle and processes (4 marks). [U]/[CO1]
12. Definition (1 mark); Causes of groundwater pollution. (3 marks); Health Hazards: Arsenicosis, Fluorosis, Blue Baby Syndrome, Water-borne pathogens. (Any 2 and explanation- 4 marks). [An]/[CO4]
13. Prevention: Afforestation, proper drainage control (4 marks); Mitigation: Retaining walls, slope leveling, hazard zonation mapping. (4 marks). [U]/[CO3]
14. Introduction (2 marks); Explanation -Key rights-right to life, freedom from torture, fair trial, freedom of expression or religion. (Any explanation -6 marks). [U]/[CO5]

[8×6=48]

Part C
Essays in 400 words
Answer any 1 question
Each question carries 12 marks

15. Introduction and definition of groundwater pollution. (2 marks); Natural factors: Rock weathering, Arsenic/Fluoride leaching (2 marks); Anthropogenic factors: Agricultural runoff, Industrial effluents, Landfill leachate, Septic systems, Mining. (Any 4 factors-8 marks). [An]/[CO4]
16. EIA-Definition (2 marks); effectiveness in environmental protection (7 marks); effectiveness in decision making (3 marks). [E]/[CO3]

[12×1=12]

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